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Interventions

All patients in the interventional group were placed in prone position immediately after randomization and received continuous prone position ventilation for a minimum of 16 h (PGE, 8.5 minutes). In the prone SCMO group, 17 patients (19.8%) discontinued the prone intervention before achievement of a prone position ventilation session (eTable 3 in Supplement 2). Most often, due to improvement in oxygenation and respiratory system compliance, the decision to discontinue prone position was made after a median (IQR) duration between prone therapy initiation and SCMO initiation of 2.0 (1.0-3.0) days (eTable 4 in Supplement 2). Prone positioning compliance, frequency, and duration were not significantly changed before and after SCMO initiation (eTable 5 in Supplement 2). Among patients randomized to the supine SCMO group, 5 (5.8%) were placed in prone position for extracorporeal membrane oxygenation (ECMO) (eTable 3 in Supplement 2).

Primary Outcome

Ventilator-free days at 28 days occurred within 65 days of randomization in 38 patients (44.2%) and 37 of 86 patients (44.9%) in the prone and supine SCMO groups, respectively (odd difference, 0.3% [95% CI, −14.9% to 15.2%], risk difference, 0.7%; subdistribution hazard ratio, 1.11 [95% CI, 0.73-1.69]; P = 0.62; eFigure 1 in Supplement 2). Ventilator-free days at 28 days were 7.0 (IQR, 0.0-19.0) in the prone group and 6.0 (IQR, 0.0-19.0) in the supine SCMO group, with no significant difference between groups (Table 2). Among patients with ECMO excluding failure, 10 were removed from SCMO (eTable 3 in Supplement 2). Among patients randomized to the supine SCMO group, 5 (5.8%) were placed in prone position for extracorporeal membrane oxygenation (ECMO) (eTable 3 in Supplement 2).

Table 2. Primary and Secondary End Points and Adverse Events[a]

Outcomes/events	Prone SCMO (n = 86)	Supine SCMO (n = 86)	Mean, median, or odd difference (95% CI)	P value
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Competing events			0.7 (−14.9 to 15.2)	
Death without ever ventilator weaning, No. (%)	35 (37.2)	31 (36.9)	0.3 (−14.9 to 15.2)	1.0
ECMO weaning failure, No. (%)	10 (10.6)	5 (5.6)	5.0 (−3.1 to 13.2)	0.34
Secondary outcomes				
Respiratory system compliance at 96 hr, mL/cm H2O			0.9 (−1.0 to 2.8)	0.36
Day 3	26 (27.0)	17 (20.2)	7.7 (−4.2 to 19.6)	0.20
Day 7	15 (16.4)	16 (19.0)	7.4 (−4.6 to 18.9)	0.23
Days alive and free from ventilator before ventilator 7 days, median (IQR)	7 (4-7)	7 (0-7)	0 (0 to 0)	.86
Days alive and free from cardiovascular failure within 7 days, median (IQR)	7 (5-7)	7 (3-7)	0 (−0.5 to 0.5)	.6
Pneumothorax by day 28, No. (%)	24 (24)	17 (21)	−0.1 (−8.7 to 8.4)	1.0
[1] Barotrauma-associated mortality, No. (%)	17 (19.5)	15 (18)	−0.4 (−12.6 to 11.8)	1.0
All-cause day 28 mortality, No. (%)	26 (37)	26 (42)	5.0 (−12.2 to 22.2)	.49
All-cause day 90 mortality, No. (%)	40 (51)	40 (48)	2.1 (−12.7 to 16.9)	.80
Days free from ECMO to day 28 (ECMO survivors), median (IQR)	19 (7-19)	7 (0-6)	10.0 (1.0 to 19.0)	.04
Serious adverse events, No. (%)	19 (23)	19 (23)	0.0 (−12.3 to 12.3)	.99
[2] Cardiac arrest	2 (2.5)	11 (13.1)	−10.6 (−19.4 to −1.8)	.02
Bleeding event requiring transfusion	24 (27.9)	19 (18.1)	9.8 (−2.1 to 21.7)	.13
Hemorrhagic stroke	2 (2.5)	1 (1.2)	1.3 (−2.9 to 5.5)	1.00
Unintentional ECMO decannulation	0	0		
Nonscheduled extubation, No. (%)	0	0		
Severe hypoxemia, No. (%)	0	0		
Maximum Revised Pressure Ulcer Staging Score, median (IQR)	3 (3-3)	3 (2-3)	2.1 (−4.0 to 8.2)	.14

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

Supplement 1.

eTable 1. Characteristics of participating sites

eTable 2. Inclusion and exclusion criteria

eTable 3. Prone positioning procedure on ECMO

eTable 4. ECMO management protocol

eTable 5. ECMO weaning

eTable 6. Patient enrollment by participating sites

eTable 7. Reasons for discontinuing PP or having < 4 PP sessions for patients randomized in the prone ECMO arm

eTable 8. Evolution of respiratory system compliance, PaO2, and PaCO2 before and after 16 hours of prone positioning on ECMO (first prone positioning session).

eTable 9. Cause-specific analysis of the primary outcome

eFigure 1. Median respiratory system compliance (interquartile range) and its 95% confidence interval from Day 1 to Day 7 post-randomization according to prone or supine position

eFigure 2. Cumulate incidence function for the events of ECMO successful weaning, weaning failure due to death after ECMO separation, weaning failure due to the need for a second ECMO run, and death, by position group

eFigure 3. Kaplan-Meier time-to-event curves for survival in the prone and supine ECMO groups

eFigure 4. Kaplan-Meier time-to-event curves for the probability of being on ECMO in the prone and supine ECMO groups

eFigure 5. Kaplan-Meier time-to-event curves for the probability of being on mechanical ventilation in the prone and supine ECMO groups

eFigure 6. Kaplan-Meier time-to-event curves for the probability of remaining in the ICU in the prone and supine ECMO groups

eFigure 7. Kaplan-Meier time-to-event curves for the probability of remaining in the hospital in the prone and supine ECMO groups

Click here for additional data file. (846.5KB, pdf)

Supplement 2.

Trial Protocol

Click here for additional data file. (1.9MB, pdf)

Supplement 3.

Nonauthor Collaborators. PRONECMO Investigators

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Supplement 4.

Data Sharing Statement

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